AuxTransUNet: Enhancing Remote Sensing Image Segmentation of Open-Pit Mining Areas in Qinghai–Tibet Plateau

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***Abstract*—Accurate and efficient extraction of mining areas from remote sensing imagery is essential for resource investigation, environmental assessment, and ecological management. This task is particularly crucial for the intelligent analysis of large-scale mining landscapes, such as those found on the Qinghai–Tibet Plateau. However, existing approaches face notable limitations. Many deep learning models, especially those based on convolutional neural networks (CNNs), struggle to capture the complex and heterogeneous morphological features of mining areas in diverse geographic settings. To address these challenges, we propose AuxTransUNet, a hybrid deep learning framework that integrates CNNs with transformers to enhance both local detail extraction and global contextual understanding. The architecture incorporates a frequency-aware and boundary-guided feature fusion strategy, which improves segmentation accuracy by suppressing misclassification noise and refining boundary delineation. In addition, an auxiliary classification branch provides patch-level supervision, further strengthening semantic consistency. Extensive experiments conducted on a mining segmentation dataset covering the Qinghai–Tibet Plateau demonstrate that AuxTransUNet achieves superior performance compared to strong baseline models in terms of both segmentation accuracy and computational efficiency. The results highlight its potential as a robust and scalable solution for large-area mining monitoring using remote sensing data.**

***Index Terms*—Convolutional neural network (CNN)–transformer hybrid, land use mapping, open-pit mining, Qinghai–Tibet plateau, remote sensing, semantic segmentation.**

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I. Introduction

MINERAL resources are fundamental to the production and development of human society [1], playing a vital role in economic, social, and national security. The Qinghai–Tibet Plateau, rich in mineral reserves, serves as a long-term strategic resource base essential for China's socioeconomic development. However, its remote location and complex terrain make it one of the least explored regions for mineral investigation, posing substantial challenges for a timely and accurate mining area assessment. As demand for resource management, inventory updates, and green mining grows, there is an urgent need for efficient methods to extract mining-related information over such vast and rugged landscapes.

Traditional mine monitoring relies predominantly on visual interpretation of remote sensing imagery combined with field validation. While accurate, this approach is labor intensive, time-consuming and not scalable for large-scale or long-term observation, especially on the Qinghai–Tibet Plateau. It also relies heavily on domain expertise, limiting its practicality for widespread implementation. Traditional remote sensing classification methods rely on low-level spectral, textural, or geometric features extracted from imagery and fed into machine learning algorithms, such as maximum likelihood classification, support vector machines, decision trees, or random forests [2], [3]. However, pixel-level classification often suffers from salt-and-pepper noise due to high intraclass variability and low interclass separability. Object-based image analysis alleviates this issue to some extent and has been widely used to extract features from high-resolution mining imagery [3], [4].

The emergence of deep learning has significantly improved the land cover mapping capabilities in remote sensing. Convolutional neural networks (CNNs), such as AlexNet [5], VGG [6], ResNet [7], and DenseNet [8], have enabled powerful representation learning. These networks have been widely adopted in mining applications. For example, Chen et al. [9] developed a CNN-based object-oriented framework to delineate open-pit regions, and [10], [11] improved feature extraction for mining and waste dump detection using ResNet and DenseNet. U-Net [12] and its extensions, including Unet++ [13], have also been applied to mining segmentation tasks in diverse geographic regions [14], [15], [16].

While CNNs excel at capturing local features, they are limited in modeling long-range dependencies, an essential capability for interpreting the complex spatial patterns of mining landscapes. To address this, transformer-based architectures have been introduced, using attention mechanisms and dynamic receptive fields for enhanced contextual modeling. Pioneering models such as detection transformer [17], vision transformer (ViT) [18], and their semantic segmentation variants, SETR [19], segmenter [20], and SegFormer [21], have achieved state-of-the-art performance in various computer vision and remote sensing tasks [22], [23], [24], [25]. In the context of mining, several recent studies have adapted transformer models to address the challenges of irregular terrain, fragmented land cover, and sparse annotations. Bi et al. [26] proposed the agent mining transformer, which dynamically mines local feature agents to improve semantic consistency in few-shot segmentation of open-pit mining regions. Similarly, Qiao et al. [27] developed SegMine, a hybrid architecture that integrates a ViT-based encoder with an attention-guided decoder to extract the boundaries of the coal mine from hyperspectral imagery. These models demonstrate that transformer-based frameworks can offer superior generalization and boundary precision in mining area extraction compared to conventional CNNs. These advances provide a strong methodological foundation for further integrating Transformer modules into task-specific architectures for mining applications.

Building on this foundation, the broader field of remote sensing semantic segmentation has seen a surge in task-specific models that leverage advanced architectures, such as transformers to tackle persistent, intertwined challenges. Key research frontiers include refining object boundaries, modeling multiscale contextual information, and integrating multimodal or frequency-aware data. A primary focus has been on improving the delineation of class boundaries, which are often ambiguous in remote sensing imagery. To address this, some methods propose novel loss functions or dedicated architectural components. For instance, Huang et al. [28] incorporated an edge optimization module into their multiscale framework to enhance boundary precision. Similarly, PV segmenter [29] introduces a frequency-guided edge-aware design that leverages high-frequency signals to improve boundary localization and object separation in complex spatial patterns. Simultaneously, effectively modeling the vast scale variations of ground objects, from small individual structures to large contiguous land cover areas, is another critical challenge. In response, recent works have focused on designing more powerful context aggregation modules. Architectures such as AFENet [30] disentangle high- and low-frequency components and apply selective fusion strategies to robustly represent both fine textures and large-scale semantics, improving adaptability across varied spatial configurations. Furthermore, the fusion of multimodal data to create richer and more discriminative feature representations remains a highly active research area. This is particularly relevant for leveraging complementary information from diverse sensors, such as optical, elevation, and SAR data. Architectures such as FTransDeepLab [31] and SiMultiF [32] exemplify this trend, using transformer-based fusion modules and adaptive weighting schemes to effectively integrate information from different modalities.

While these state-of-the-art approaches individually target critical aspects, such as boundaries, scale, and multimodality, significant challenges remain when deploying them in uniquely complex geospatial environments, such as the Qinghai–Tibet Plateau. The mining areas of this region, characterized by a combination of irregular morphology, fuzzy class transitions, and heterogeneous textures, demand a holistic solution. Even advanced models can produce oversmoothed predictions or fail to accurately delineate intricate features, reducing their utility for practical ecological assessments. This underscores the need for a novel, hybrid architecture that not only integrates these advanced concepts but is also custom-tailored to the distinct spectral and spatial characteristics of mining landscapes.

To address these challenges, we propose AuxTransUNet, a novel hybrid network that combines the strengths of CNNs and ViTs for robust segmentation of mining areas. At its core, AuxTransUNet features a dual-branch encoder: one branch focuses on multispectral feature extraction to capture spectral diversity, while the other emphasizes frequency-domain representations to better preserve fine-scale details, such as irregular mining boundaries. This structure is complemented by an auxiliary classification branch, which guides the network in learning more discriminative global representations through multitask learning. In addition, we incorporate a boundary-aware loss to improve edge localization and reduce segmentation ambiguity. Extensive experiments conducted on a curated dataset of open-pit mining areas from the Qinghai–Tibet Plateau show that AuxTransUNet consistently outperforms representative baseline models in terms of segmentation accuracy. The main contributions of this work are as follows.

1) AuxTransUNet is proposed as a task-adaptive hybrid architecture that integrates CNN and transformer modules for effective segmentation of large-scale heterogeneous mining areas.
2) A dual branch encoder and an auxiliary classification mechanism are introduced to enhance global–local feature representation and improve segmentation performance under limited supervision.
3) A boundary-aware loss function is incorporated to improve edge precision, addressing the frequent challenge of boundary ambiguity in mining area segmentation.

II. Study Area and Materials

This study focuses on mining sites distributed across the Qinghai–Tibet Plateau, a geologically and ecologically sensitive region characterized by high elevation, sparse vegetation, and complex terrain. Vector point data delineating mining site locations were sourced from official government and reliable databases, while polygon boundary data for select sites were extracted from previously published research [33], [34], [35]. To ensure precise spatial referencing, manual interpretation was conducted using high-resolution imagery available on Google Earth, enabling the accurate delineation of open-pit mining extents.

To ensure semantic consistency during annotation, the category of “open-pit mining area” was defined to include all

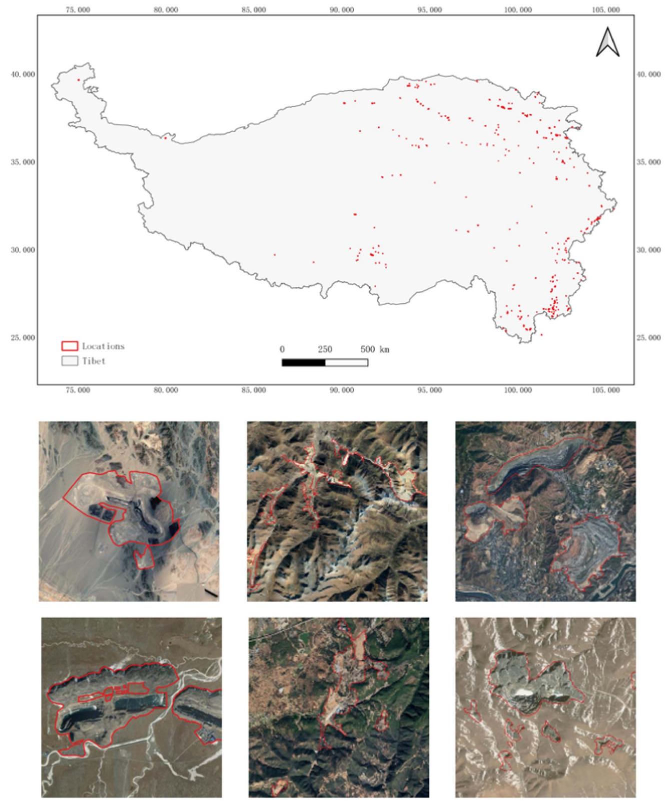

Fig. 1. Distribution of mining samples across the Qinghai–Tibet Plateau and some examples of labels.

functional zones directly associated with surface mining operations, such as excavated pits, dumping grounds, ore stockpiles, settling ponds, tailing storage facilities, and associated internal infrastructure (e.g., roads and buildings) within or connected to the mining footprint. Conversely, isolated roads and buildings outside these zones, where the connection to mining operations could not be confidently established, were excluded from annotation to avoid semantic ambiguity. The final spatial distribution of samples and representative examples are illustrated in Fig. 1.

The remote sensing data used in this study consist of high-resolution imagery captured by the Sentinel-2 satellite, which is part of the European Space Agency's Copernicus program. Sentinel-2 is equipped with a multispectral instrument that provides 12 spectral bands with spatial resolutions of 10, 20, and 60 m, covering the visible, near-infrared, and shortwave infrared (SWIR) regions. These attributes make Sentinel-2 particularly suitable for environmental monitoring, land cover classification, and other detailed surface analyses. The Sentinel-2 Level-2 A product, which includes atmospheric correction, was selected to ensure accurate surface reflectance measurements. The selected dataset corresponds to the growing season from May to October in 2020, a period chosen to reduce the impact of snow cover and seasonal variability while maximizing visibility of vegetation and mining activity, both critical for open-pit mining segmentation. To ensure data quality and usability, rigorous preprocessing steps were applied. Cloud contamination, a common issue in remote sensing, is especially prominent in high-altitude regions. Cloud masking was performed using the built-in quality flags of the Sentinel-2 Level-2 A product to remove affected pixels. In addition, non-valid observations were discarded, and mean compositing was applied to valid observations to maintain temporal consistency and minimize noise in cloud-affected areas.

III. Method

A. Overall Architecture

We propose AuxTransUNet, a hybrid encoder–decoder framework that integrates convolutional and transformer-based designs for accurate semantic segmentation in complex mining areas. The overall architecture is illustrated in Fig. 2 and consists of five main components: a hybrid encoder, a ViT backbone, a hierarchical decoder, a patch-level auxiliary supervision head, and a composite loss function.

The encoder combines spatial features extracted by a ResNet backbone with frequency-aware information derived from a dedicated high-frequency extractor. These features are fused and tokenized as inputs to a ViT encoder, which models long-range dependencies across the image. Multiscale skip connections are retained for subsequent fusion in the decoder. The decoder gradually upsamples transformer features with skip connections from the convolutional backbone to recover spatial resolution. Segmentation is performed through a lightweight convolutional head. To enhance training supervision and global semantic understanding, an auxiliary classification head is attached at the patch level, operating on selected transformer block outputs. This hybrid design effectively leverages spatial precision, global context, and frequency-domain priors. Detailed descriptions of each module are provided in Sections B–D.

B. Frequency-Aware Feature Extraction

To enhance the model's sensitivity to fine-grained textures and irregular mining boundaries, we introduce a frequency-aware feature extractor that captures complementary high-frequency information from the input image. The module [see Fig. 2(a)] transforms the input image $\mathbf{I} \in \mathbb{R}^{B\times C\times H\times W}$ into the frequency domain using a 2-D Discrete Fourier Transform (DFT)

$$\mathcal{F}(\mathbf{I}) = \hat{\mathbf{I}} = \text{DFT}(\mathbf{I}) = \hat{\mathbf{I}}_{\text{real}} + j\hat{\mathbf{I}}_{\text{imaginary}} \tag{1}$$

where $\hat{\mathbf{I}}_{\text{real}}$ and $\hat{\mathbf{I}}_{\text{imaginary}}$ denote the real and imaginary parts of the frequency components, respectively.

To isolate informative high-frequency components (e.g., edges and textures), we construct a frequency mask $M(u,v)$ based on the radial frequency magnitude

$$M(u,v) = \begin{cases} 1, \text{if } \sqrt{u^2+v^2} > \theta \\ 0, \text{otherwise} \end{cases} \tag{2}$$

where θ is a threshold empirically set to filter out low-frequency components that contribute less to boundary localization.

Next, both masked components are processed through two shared convolutional pathways to extract semantic frequency-aware features. The resulting frequency feature map is formed

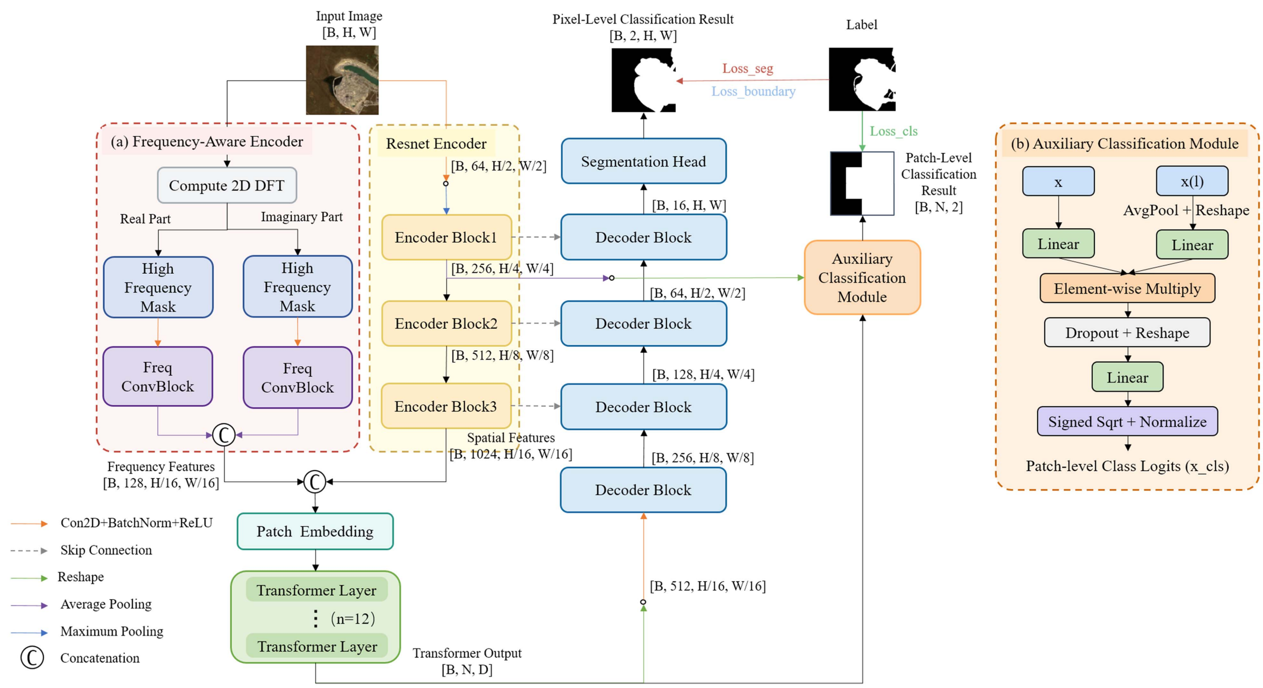

Fig. 2. Overall architecture of the AuxTransUNet network. It includes (a) frequency-aware encoder and (b) auxiliary classification module.

via concatenation

$$\mathcal{F}_{\text{freq}} = \text{Concat}\left(\phi_{\text{real}}(M \cdot \hat{\mathbf{I}}_{\text{real}}),\ \phi_{\text{imaginary}}(M \cdot \hat{\mathbf{I}}_{\text{imaginary}})\right). \tag{3}$$

Finally, the frequency features are concatenated with the output from the spatial encoder (ResNet) to form a hybrid embedding

$$\mathcal{F}_{\text{emb}} = \text{Concat}(\mathcal{F}_{\text{resnet}},\ \mathcal{F}_{\text{freq}}). \tag{4}$$

This fusion enables the network to leverage both spatial and frequency cues, thereby improving its capability to delineate complex mining geometries and boundary details that are often missed in the spatial domain alone.

C. Auxiliary Patch-Level Supervision for Enhanced Segmentation

To improve the discriminative power of the model in complex and heterogeneous landscapes, we introduce an auxiliary patch-level supervision mechanism that serves as a complementary objective to the main segmentation head. Unlike conventional classification branches that predict global image categories; this module operates at the patch level, encouraging the model to learn semantically meaningful representations across local regions while maintaining consistency with global semantics.

The design is grounded in the observation that mining areas often exhibit mixed land-cover features and ambiguous boundaries, which challenge pixelwise segmentation alone. By assigning semantic labels to individual patches, the auxiliary branch captures high-level structural patterns and long-range dependencies, thus providing global context to guide the local predictions made by the segmentation head. The auxiliary classification structure is illustrated in Fig. 2(b).

Technically, feature embeddings from both the final transformer output $\mathbf{x}$ and an earlier ResNet encoder block $\mathbf{x}^{(l)}$ are first projected into a shared high-dimensional space

$$\mathbf{z}_1 = \mathbf{W}_1\mathbf{x}, \quad \mathbf{z}_2 = \mathbf{W}_2\mathbf{x}^{(l)} \tag{5}$$

where $\mathbf{W}_1$ and $\mathbf{W}_2$ are learnable projection matrices.

Then, a bilinear fusion strategy is applied to the projected features to capture cross-level semantic interactions

$$\mathbf{z} = \text{Dropout}(\mathbf{z}_1 \odot \mathbf{z}_2) \tag{6}$$

where $\odot$ denotes elementwise multiplication.

The resulting fused tensor is reshaped to form patchwise classification vectors, which are subsequently passed through a fully connected layer

$$\mathbf{y} = \text{Linear}(\text{Reshape}(\mathbf{z})). \tag{7}$$

Finally, the output logits are passed through a signed square-root operation followed by ℓ_2 normalization to enhance numerical stability and classification consistency

$$\text{logits}_{\text{cls}} = \text{Normalize}\left(\text{sign}(\mathbf{y}) \cdot \sqrt{|\mathbf{y}| + \epsilon}\right). \tag{8}$$

Overall, this auxiliary module offers additional supervision to guide the encoder toward learning semantically consistent representations. It improves the model's robustness to spatial ambiguity and heterogeneous land-cover conditions, ultimately enhancing segmentation accuracy and generalization performance.

D. Loss Function

To supervise the model more effectively during training, the total loss function in AuxTransUNet integrates three components: segmentation loss, auxiliary classification loss, and boundary loss (BL).

Both the segmentation loss and the auxiliary classification loss adopt a composite form that combines cross-entropy and dice losses, balancing per-pixel accuracy and overlap-based region alignment. While the segmentation loss supervises the pixelwise prediction, the auxiliary classification loss guides patch-level predictions derived from intermediate transformer features. These two losses are formulated as

$$\mathcal{L}_{\text{seg}} = \mathcal{L}_{\text{ce}}^{\text{seg}} + \mathcal{L}_{\text{dice}}^{\text{seg}} \tag{9}$$

$$\mathcal{L}_{\text{cls}} = \mathcal{L}_{\text{ce}}^{\text{cls}} + \mathcal{L}_{\text{dice}}^{\text{cls}}. \tag{10}$$

To further refine boundary delineation, we introduce a level set-based BL, which minimizes the energy of the predicted probability field relative to the ground truth. Given the predicted map $\mathbf{P}$ and one-hot encoded ground truth $\mathbf{G}$, the BL is computed as

$$\mathcal{L}_{\text{boundary}} = \alpha \cdot \left[(1-\mathbf{P})^2 \cdot \mathbf{G} + \mathbf{P}^2 \cdot (1-\mathbf{G})\right] \tag{11}$$

where α is a weighting factor that controls the influence of this term. This design enhances the model's sensitivity to edge structures, particularly useful for delineating complex mining boundaries.

The total loss function is the weighted sum of all three components

$$\mathcal{L}_{\text{total}} = \mathcal{L}_{\text{seg}} + \mathcal{L}_{\text{cls}} + \mathcal{L}_{\text{boundary}}. \tag{12}$$

This formulation ensures that the network simultaneously learns accurate pixelwise classification, global context-aware supervision, and precise boundary localization, leading to enhanced segmentation performance, especially in heterogeneous and mixed land-use mining environments.

IV. Experiment Results and Analysis

A. Experimental Settings and Evaluation Metrics

1) Dataset and Implementation Details: The dataset used in this study consists of open-pit mining samples extracted from the Qinghai–Tibet Plateau. A total of 1248 mining sites were manually delineated on high-resolution basemaps. Each site was clipped into patches of size 256×256 pixels from corresponding Sentinel-2 imagery. After excluding patches without discernible mining features, a total of 2429 valid image samples were retained. These were randomly split into 1700 training samples (70%), 485 validation samples (20%), and 244 test samples (10%).

All experiments were conducted using PyTorch on a single NVIDIA GeForce RTX 3090 GPU with 24 GB of memory. Models were trained for 150 epochs using the AdamW optimizer with default settings and an initial learning rate of 0.0001, decayed by a factor of 0.1 every 50 epochs. The input images were resized to 256×256 pixels, and the batch size was set to 16. Data augmentation techniques, including random rotation and horizontal flipping, were applied on-the-fly during training to improve model robustness.

2) Evaluation Metrics: To comprehensively evaluate model performance, we adopted four commonly used metrics: mean intersection over union (mIoU), overall accuracy (OA), Kappa coefficient, and F1 score. These metrics assess region-level overlap, classification accuracy, interrater agreement, and the balance between precision and recall. The formulas are defined as follows:

$$\text{mIoU} = \frac{1}{N}\sum_{i=1}^{N}\frac{A_i \cap B_i}{A_i \cup B_i} \tag{13}$$

$$\text{OA} = \frac{\text{TP}+\text{TN}}{\text{TP}+\text{TN}+\text{FP}+\text{FN}} \tag{14}$$

$$\kappa = \frac{p_o - p_e}{1 - p_e} \tag{15}$$

$$\text{Recall} = \frac{\text{TP}}{\text{TP}+\text{FN}} \tag{16}$$

$$\text{Precision} = \frac{\text{TP}}{\text{TP}+\text{FP}} \tag{17}$$

$$F1 = \frac{2\text{TP}}{2\text{TP}+\text{FN}+\text{FP}} = \frac{2\cdot\text{Precision}\cdot\text{Recall}}{\text{Precision}+\text{Recall}}. \tag{18}$$

B. Quantitative and Qualitative Evaluation of Segmentation Performance

This section presents the experimental results comparing the performance of different semantic segmentation networks for mining area extraction on the Qinghai–Tibet Plateau. The experiments were carried out using a variety of network architectures, including U-Net, DeepLab v3, DeepLab v3+, TransUNet, Swin-Unet, and SegFormer. All networks were tested using identical spectral band configurations to ensure consistent evaluation conditions. The detailed quantitative results are summarized in Table I.

From the comparison, it is evident that our proposed AuxTransUNet consistently outperforms all other benchmark models in terms of segmentation accuracy. Specifically, AuxTransUNet achieves the highest mIoU of 0.9216 and OA of 0.9841, representing a 2.17% and 0.48% improvement over SegFormer, respectively. Compared to TransUNet, it yields a 2.35% gain in mIoU and a 0.5% increase in OA, along with the highest F1 and Kappa scores across all models. These results validate the effectiveness of our design in capturing complex semantic structures in open-pit mining areas.

Although models such as U-Net and DeepLab v3+ exhibit higher inference speeds or have fewer parameters, they fall short in segmentation performance. Although AuxTransUNet is slightly more computationally intensive than some CNN-based models (108.72 M parameters and 36.49 GFLOPs), it maintains a competitive inference speed of 32.12 FPS, which is comparable to SegFormer (34.02 FPS) and TransUNet (33.89 FPS). This demonstrates a well-balanced tradeoff between accuracy and computational efficiency, making AuxTransUNet suitable for practical applications that demand both high precision and reasonable inference speed.

TABLE I
COMPARISON OF SEMANTIC SEGMENTATION FRAMEWORKS ON MINING AREA DATASET

Methods	mIoU	OA	Kappa	F1	Params (M)	GFLOPs	FPS
U-Net [12]	0.8586	0.9683	0.8411	0.9206	4.32	10.23	273.32
DeepLab v3 (ResNet50) [36]	0.8625	0.9698	0.8456	0.9228	39.66	12.53	97.44
DeepLab v3+ (ResNet50) [37]	0.8777	0.9737	0.8645	0.9323	39.78	15.30	95.30
ViT-Unet	0.8216	0.9608	0.7919	0.8959	95.72	30.64	72.17
R50-ViT-Unet	0.8494	0.9674	0.8287	0.9143	106.46	30.39	34.09
TransUNet [38]	0.8981	0.9791	0.8890	0.9445	107.97	33.41	33.89
Swin-Unet [39]	0.8401	0.9657	0.8166	0.9083	27.19	7.77	49.81
SegFormer [21]	0.8999	0.9793	0.8912	0.9456	109.08	33.41	34.02
Ours	**0.9216**	**0.9841**	**0.9163**	**0.9581**	**108.72**	**36.49**	**32.12**

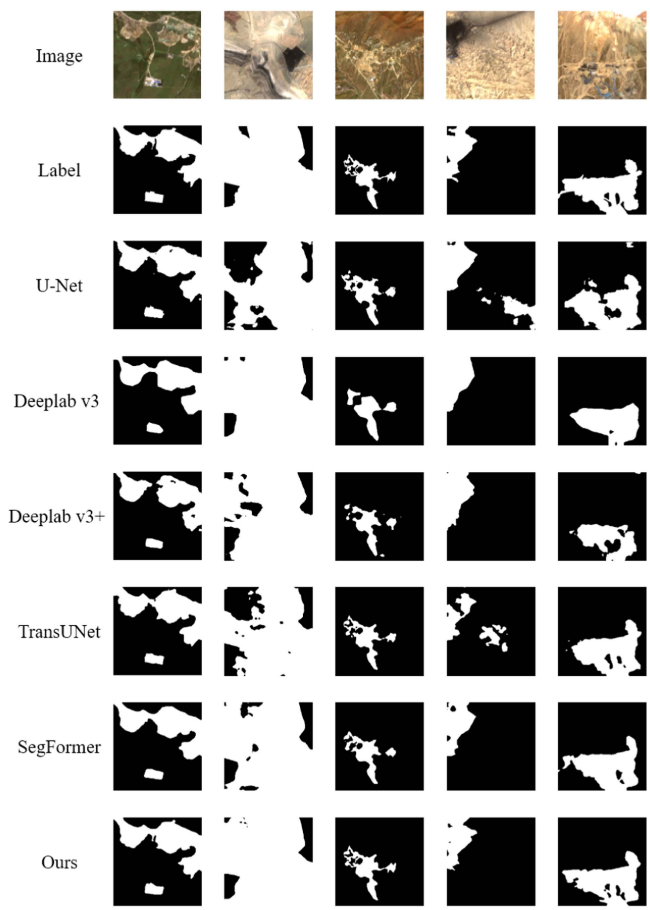

Fig. 3. Visualization of segmentation results from different networks on mining area dataset.

Furthermore, an interesting trend was observed when analyzing the performance of U-Net and its variants with different backbones, such as ViT, ResNet50, and Swin Transformer, where skip connections were omitted. These variants did not outperform the original U-Net, emphasizing the importance of multiresolution feature integration within the decoder. This result highlights the critical role of multiscale feature fusion in maintaining high segmentation performance, particularly in complex environments, such as open-pit mining areas.

Fig. 3 presents a comprehensive visual comparison of the segmentation results produced by various networks on the mining area dataset. The mining regions in question exhibit considerable heterogeneity in morphology, with intricate internal structures and ambiguous boundary definitions that pose significant challenges for automated extraction. As observed, transformer-based methods consistently outperform traditional CNN-based architectures in capturing these complex spatial patterns. Among them, AuxTransUNet yields the most accurate delineation of mining boundaries while substantially reducing both misclassification and omission errors. These results underscore the effectiveness of combining global context modeling from transformers with the fine-grained localization capabilities of CNNs to segment large-scale and topographically diverse mining regions.

TABLE II
IMPACT OF CLASSIFICATION INPUT MODALITIES ON SEGMENTATION PERFORMANCE

Input Stages	mIoU	OA	Kappa	F1
None	0.8775	0.9742	0.8642	0.9321
[4]	0.8799	0.9746	0.8671	0.9336
[2, 4]	**0.8884**	**0.9769**	**0.8774**	**0.9387**
[2, 3, 4]	0.8860	0.9759	0.8746	0.9373
[1, 2, 3, 4]	0.8812	0.9747	0.8687	0.9343

Fig. 4 further illustrates the segmentation outputs of three representative sites: the Muli Coalfield, the Yulong Copper Mine, and the Tanjianshan Gold Mine. Red boundary boxes highlight areas where competing models exhibit significant misclassifications or boundary inaccuracies. In contrast, AuxTransUNet demonstrates superior fidelity in boundary preservation and a lower incidence of false predictions. This enhanced performance highlights the robustness of the model in identifying spatially complex mining features and affirms its suitability for practical applications in large-area mining monitoring and environmental assessment.

C. Performance Impact of Varying Auxiliary Classification Inputs

To investigate the effect of different combinations of feature input for the auxiliary classification module, we evaluated multiple configurations derived from various stages of the backbone encoder. Our backbone consists of four hierarchical stages, each capturing features at different levels of abstraction. The performance of five configurations, ranging from no classification input to full multistage fusion, is summarized in Table II.

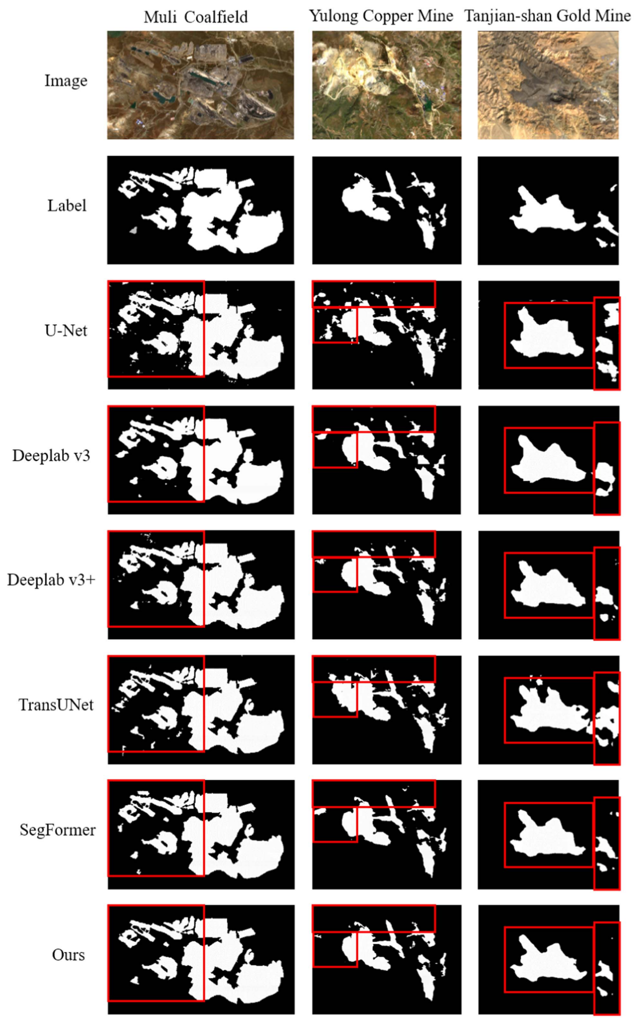

Fig. 4. Comparative visualization of segmentation results from different models in mining areas of the Qinghai–Tibet Plateau.

The configuration without any classification branch served as the baseline, achieving moderate performance with a mIoU of 0.8775, OA of 0.9742, Kappa coefficient of 0.8642, and F1 score of 0.9321. Introducing stage 4 features alone led to a slight improvement (mIoU: 0.8799, Kappa: 0.8671); however, this setting relied solely on high-level semantic information, which lacked sufficient spatial detail for significant gains.

In contrast, the configuration combining stage 2 and stage 4 features resulted in the best overall performance, with a mIoU of 0.8884, OA of 0.9769, Kappa of 0.8774, and F1 score of 0.9387. This suggests that combining mid-level (spatially detailed) and high-level (semantically rich) features enables the network to better balance spatial precision and contextual understanding, ultimately enhancing segmentation quality.

Adding stage 3 to the [2, 4] configuration (that is, [2, 3, 4]) resulted in slight improvements in mIoU and F1 score. However, the performance gains were outweighed by the added model complexity, suggesting limited cost-effectiveness of further deepening the feature aggregation structure. In particular, incorporating features from all four stages ([1, 2, 3, 4]) led to a decrease in performance, with mIoU dropping to 0.8812. This reduction may be attributed to the introduction of redundant or noisy low-level features from stage 1, which confused the classifier and diluted the effectiveness of more informative feature maps.

Overall, these results indicate that the optimal input modality for the classification branch is the selective fusion of features from stages 2 and 4, offering a favorable tradeoff between accuracy and model complexity.

TABLE III
ABLATION STUDY OF KEY COMPONENTS

ACM	FEFM	BL	mIoU	OA	Kappa	F1
			0.8981	0.9791	0.8890	0.9445
✓			0.9103	0.9816	0.9034	0.9517
	✓		0.9082	0.9813	0.9010	0.9504
		✓	0.9127	0.9822	0.9061	0.9530
✓	✓		0.9141	0.9825	0.9077	0.9539
✓		✓	0.9145	0.9824	0.9081	0.9541
✓	✓	✓	**0.9216**	**0.9841**	**0.9163**	**0.9581**

D. *Componentwise Ablation Study*

To assess the individual contributions of critical components within AuxTransUNet, we conducted a series of ablation experiments. The modules under investigation include the auxiliary classification module (ACM), the frequency-enhanced feature module (FEFM), and the BL. By systematically adding or removing these components, we assessed their respective impacts on segmentation performance. The results are summarized in Table III.

The baseline model, without ACM, FEFM, or BL, achieved an mIoU of 0.8981 and a Kappa coefficient of 0.8890. Introducing ACM led to a substantial improvement, raising the mIoU to 0.9103 and the Kappa to 0.9034. These gains demonstrate the effectiveness of auxiliary patch-level supervision in reinforcing semantic consistency and improving the model's ability to capture large-scale spatial structures, particularly in complex mining scenarios.

Similarly, the inclusion of FEFM improved both mIoU (from 0.8981 to 0.9082) and Kappa (to 0.9010), confirming its utility in extracting frequency domain features that complement spatial representations. This module proves especially beneficial for capturing fine-grained textures and boundaries typical of heterogeneous mining landscapes.

Among the individual components, BL yielded the largest single gain. By enforcing a tighter alignment between predicted boundaries and ground truth edges, BL raised the F1 score from 0.9445 to 0.9530 and improved the mIoU to 0.9127.

When modules were merged, synergistic effects were observed. The joint use of ACM and FEFM led to improved segmentation (mIoU: 0.9141), and the addition of BL to either module further elevated accuracy. The complete model, which integrates ACM, FEFM, and BL, achieved the highest performance in all metrics, including an mIoU of 0.9216, an OA of 0.9841, and an F1 score of 0.9581.

These results highlight the complementary roles of the three components: ACM provides semantic reinforcement through

TABLE IV
EFFECT OF LOSS WEIGHTS ON SEGMENTATION PERFORMANCE

λ_1	λ_2	BL	FEFM	mIoU	OA	Kappa	F1
0.5	0.5			0.9103	0.9816	0.9034	0.9517
1.0	0.5			0.9111	0.9817	0.9042	0.9521
0.5	1.0			0.9103	0.9816	0.9033	0.9517
1.0	1.0			0.9121	0.9820	0.9054	0.9527
0.5	0.5	✓		0.9080	0.9810	0.9007	0.9505
1.0	0.5	✓		0.9052	0.9804	0.8974	0.9487
0.5	1.0	✓		0.9100	0.9815	0.9030	0.9515
1.0	1.0	✓		0.9145	0.9824	0.9081	0.9541
0.5	0.5		✓	0.9092	0.9814	0.9020	0.9510
1.0	0.5		✓	0.9100	0.9815	0.9030	0.9515
0.5	1.0		✓	0.9126	0.9821	0.9060	0.9530
1.0	1.0		✓	0.9141	0.9825	0.9077	0.9539

TABLE V
EFFECT OF SPECTRAL BAND CONFIGURATION AND CLAHE PROCESSING ON SEGMENTATION PERFORMANCE

Input Bands	Processing Method	mIoU	OA	Kappa	F1
Four bands	Default	0.8683	0.9724	0.8528	0.9264
	CLAHE on RGB	0.8779	0.9746	0.8646	0.9323
	+ CLAHE RGB	0.8595	0.9701	0.8417	0.9209
Ten bands	Default	0.8775	0.9742	0.8642	0.9321
	CLAHE on RGB	0.8776	0.9743	0.8643	0.9321
	+ CLAHE RGB	0.8746	0.9735	0.8605	0.9303
Twelve bands	Default	0.8758	0.9737	0.8620	0.9310
	CLAHE on RGB	0.8626	0.9709	0.8456	0.9228
	+ CLAHE RGB	0.8606	0.9701	0.8431	0.9216
Thirteen bands	Default	0.8770	0.9741	0.8636	0.9318
	CLAHE on RGB	0.8705	0.9728	0.8555	0.9277
	+ CLAHE RGB	0.8790	0.9744	0.8660	0.9330

auxiliary classification, FEFM enriches feature representations by incorporating frequency-sensitive information, and BL enhances structural precision through boundary supervision. Their integration significantly improves the segmentation capacity of the proposed network, especially in delineating complex and irregular mining areas.

E. Impact of Loss Weighting Strategies

To evaluate the influence of different supervisory signals and their relative importance during training, we conducted a series of controlled experiments by adjusting the weights of the segmentation loss (λ_1) and the auxiliary classification loss (λ_2). In addition, we examined their interactions with the integration of BL and FEFM. The corresponding results are summarized in Table IV.

The results show that segmentation performance is sensitive to the balance between the loss components. Increasing the weight of classification loss (λ_2) generally led to improvements in Kappa and F1 scores, indicating that stronger auxiliary supervision improves the model's ability to learn global semantic patterns. The configuration with equal weights ($\lambda_1 = 1.0$, $\lambda_2 = 1.0$) yielded favorable performance in all metrics, with mIoU reaching 0.9121 and OA improving to 0.9820.

Adding BL to this balanced configuration further improved segmentation precision. By explicitly guiding the network to align the predicted and actual boundaries, the inclusion of BL increased the mIoU to 0.9145 and increased the F1 score to 0.9541, demonstrating its effectiveness in refining edge localization and structural clarity.

Furthermore, incorporating FEFM contributed to additional performance gains by capturing frequency-domain cues that enrich spatial representations. In particular, the combination of BL and FEFM under balanced loss weights ($\lambda_1 = \lambda_2 = 1.0$) produced the best overall results, achieving an mIoU of 0.9141, OA of 0.9825 and an F1 score of 0.9539. This setup demonstrates an optimal tradeoff between semantic coherence, fine-grained texture extraction, and boundary precision.

In conclusion, coordinated optimization of segmentation, classification and boundary-based objectives significantly strengthens the network's capacity to capture both semantic structure and fine boundary detail in mining area segmentation.

F. Effect of Spectral Band Selection and Contrast Enhancement

To assess how spectral band selection and preprocessing influence segmentation performance, we conducted controlled experiments using four distinct input schemes as follows.

1) *Four-band configuration:* Red, green, blue, and near-infrared (RGB+NIR), all resampled to 10 m spatial resolution.
2) *Ten-band configuration:* The above-mentioned four bands plus six additional spectral bands (originally at 20 m resolution), commonly utilized in vegetation and land cover analysis.
3) *Twelve-band configuration:* The first 12 bands of Sentinel-2 Level-2 A (SR) data, covering visible, NIR, and SWIR ranges.
4) *Thirteen-band configuration:* An extension of the 12-band input with an additional scene classification layer included.

We further evaluated the impact of applying contrast limited adaptive histogram equalization (CLAHE) to enhance local contrast in RGB channels. Two variants were tested: applying CLAHE directly to RGB bands ("CLAHE RGB") and concatenating the CLAHE-processed RGB bands to the original input ("+CLAHE RGB"). The experimental results are summarized in Table V

The ten-band configuration consistently yielded the highest and most stable performance, with an mIoU of 0.8775 and OA of 0.9742 under default preprocessing. In contrast, the 13-band configuration, which included a coarse scene classification map, did not yield further improvements and, in some cases, slightly reduced mIoU, suggesting that coarse-resolution semantic priors may introduce redundancy or compromise spatial precision.

CLAHE preprocessing was particularly beneficial in low-spectral-band scenarios. For the four-band input, applying CLAHE improved mIoU by nearly 1% (from 0.8683 to 0.8779) and increased the F1 score from 0.9264 to 0.9323. This enhancement highlights CLAHE's ability to accentuate boundaries and improve contrast where spectral diversity is limited. However, for richer spectral configurations (12 or 13 band), the benefits of CLAHE diminished and occasionally turned negative, likely due to overenhancement or distortion of spectral signatures.

Interestingly, the CLAHE-enhanced four-band input performed similarly to the more spectrally comprehensive 10- and 12-band inputs, indicating that targeted contrast enhancement can partially compensate for reduced spectral dimensionality. In contrast, the "+CLAHE RGB" variant, which concatenates the enhanced bands with the original input, often degraded performance, probably due to excessive low-level redundancy confusing the model.

In summary, optimal segmentation performance was achieved using either the ten-band configuration with default preprocessing or the four-band input with CLAHE enhancement. These findings suggest that compact yet informative spectral inputs, when combined with appropriate preprocessing, can serve as computationally efficient alternatives to full-spectrum configurations in open-pit mine segmentation tasks.

V. Discussion

A. Practical Applications and Potential

As demonstrated by our experimental results and analyses, AuxTransUNet delivers consistently high segmentation accuracy in remote sensing imagery of mining areas on the Qinghai–Tibet Plateau. Notably, our analysis of spectral band configurations reveals that using only RGB+NIR bands, when paired with effective preprocessing such as CLAHE, can deliver segmentation accuracy comparable to full-band inputs, while significantly reducing data acquisition and storage demands. This strategy streamlines preprocessing and model deployment, reduces computational overhead, and enhances usability in resource-constrained environments.

Furthermore, the Qinghai–Tibet Plateau presents highly diverse mineral resources and substantial environmental heterogeneity. AuxTransUNet's robust performance in this geologically and ecologically complex region indicates strong adaptability. This capability underscores the model's potential for generalization across varied geographic contexts and mining types, offering practical value for operational monitoring and large-scale ecological assessment.

B. Limitations and Future Directions

Although AuxTransUNet demonstrates promising segmentation performance for open-pit mining areas, several limitations remain, pointing to key avenues for future research.

From a geographic perspective, the training dataset was constructed from the Qinghai–Tibet Plateau, a region marked by unique geological and ecological characteristics. This raises concerns about generalizability of the model to mining areas with different environmental and operational conditions. Expanding the dataset to include more geographically diverse mining sites is essential to improve robustness and external validity.

Temporally, the current framework has not been evaluated using multitemporal imagery. Since mining activities and surrounding land surfaces evolve due to seasonal, climatic, and anthropogenic factors, assessing the stability of the model over time is critical for dynamic monitoring applications.

In terms of spectral generalization, although preliminary tests using reduced spectral configurations (e.g., RGB + NIR) have shown promising results, the robustness of the model to variations in sensor types and image quality remains underexplored. More evaluations are needed under diverse acquisition conditions and sensor modalities to ensure consistent performance.

Another limitation lies in the model's exclusive reliance on image data. AuxTransUNet does not incorporate domain-specific contextual information, such as geological maps, legal boundaries, or infrastructure data, which may limit the interpretability and practical decision-making value of its outputs. Integrating such external knowledge could enhance the semantic richness and real-world applicability of the model.

Finally, differences in spatial resolution and semantic label definitions across datasets pose significant challenges to model transferability. While this study focuses on Sentinel-2 imagery with a unified annotation scheme, applying the model to high-resolution datasets with differing label taxonomies may result in domain shifts and degraded performance. Addressing this issue will require strategies such as domain adaptation, label harmonization, and fine-tuning on external data.

Looking ahead, several research directions are worth pursuing. Incorporating multisource remote sensing data, such as LiDAR, hyperspectral imagery, and InSAR, could provide richer spatial and spectral representations to enhance feature extraction. Multitemporal datasets should also be introduced to enable temporal modeling and facilitate time-series tracking of mining activity and land-use transitions.

Moreover, future work should focus on validating the model across diverse ecological zones and mining types, ensuring that its strong performance is not limited to specific geographies. Finally, embedding external domain knowledge into the model framework can enhance both its interpretability and its practical utility as a decision support tool for sustainable resource management and environmental monitoring.

VI. Conclusion

This study presents AuxTransUNet, a novel semantic segmentation framework designed to delineate open-pit mining areas, particularly within the complex and heterogeneous environments of the Qinghai–Tibet Plateau. The model integrates a CNN backbone with transformer encoders to jointly capture fine-grained spatial details and global semantic context. In addition, it incorporates frequency-domain representations and auxiliary patch-level classification to enhance boundary localization and contextual understanding. To optimize learning across multiple semantic scales, a compound loss function is employed to jointly supervise pixel-level segmentation, patch-level classification, and boundary detection. This multiobjective design strengthens the model's ability to recognize structurally diverse mining patterns and improves delineation accuracy in regions with sparse or ambiguous textures.

Extensive experiments on a large-scale Sentinel-2 mining dataset demonstrate that AuxTransUNet consistently outperforms representative baselines, such as TransUNet and DeepLab v3+. The model also exhibits strong generalization across various mining types, including metallic, nonmetallic, and energy resource sites, highlighting its robustness for real-world remote sensing applications. Overall, AuxTransUNet provides a practical and scalable solution for mining-related land cover mapping and ecological monitoring. Future work will focus on enhancing generalization through multimodal data fusion and cross-regional training to support broader deployment across diverse mining landscapes globally.

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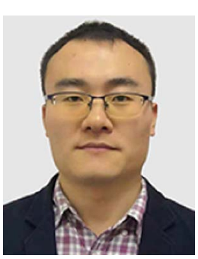

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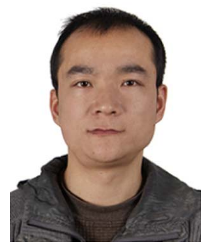

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